



1. Match List-I with List-II

List-I		List-II	
A.	Critical Region	I.	Circular wait
B.	Working Set	II.	Condition variable
C.	Deadlock	III.	Principle of locality
D.	Wait/signal	IV.	Mutual exclusion

Choose the correct answer from the options given below

- (a) A-IV, B-III, C-I, D-II (b) A-IV, B-III, C-II, D-I
(c) A-III, B-IV, C-II, D-I (d) A-III, B-IV, C-I, D-II

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2. The current allocation and maximum requirement of different types of resources for four processes are given below:

Process	Max			Allocation			Available		
	R1	R2	R3	R1	R2	R3	R1	R2	R3
P ₁	8	6	4	1	2	1	4	4	5
P ₂	4	3	3	3	1	1			
P ₃	10	1	3	4	1	3			
P ₄	3	3	3	3	2	2			

Consider the following four statements

- (A) P₂-->P₄-->P₁-->P₃ is a safe sequence.
(B) P₄-->P₂-->P₁-->P₃ is a safe sequence.
(C) P₄-->P₂-->P₃-->P₁ is a safe sequence.
(D) P₁-->P₄-->P₂-->P₃ is a safe sequence.

Identify correct statements from the given options.

- (a) A, B and D only (b) A, B and C only
(c) B, C and D only (d) A, B, C & D

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3. An operating system contains 4 user processes each requiring 5 units of resource R. The minimum number of required units of R such that no deadlock will ever occur is

- (a) 20 (b) 4 (c) 17 (d) 15

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4. How many child processes will be created by following fork () system call?

fork();
fork();
fork();
fork();

- (a) 4 (b) 16 (c) 15 (d) 3

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5. In a computer if the page fault service time is 10 ms and average memory access time is 30 ns. If one page fault is generated for every 10⁶ memory accesses. What is the effective access time for the memory?

- (a) 21 ns approximate (b) 25 ns approximate
(c) 30 ns approximate (d) 40 ns approximate

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6. Consider a system with 1K pages and 512 frames and each page is of size 2KB. How many bits are required to represent the virtual address space memory.

- (a) 20 bits (b) 21 bits (c) 11 bits (d) 16 bits

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7. We can say that a schedule is conflict serializabale?

- (a) If a schedule T can be transformed into a serial schedule U by a series of swaps of conflicting operations.
(b) If a schedule T can be transformed into a serial schedule U by a series of swaps of nonconflicting operations.

- (c) If a schedule T can be transformed into a nonserial schedule U by a series of swaps of conflicting operations.
(d) If a schedule T can be transformed into a nonserial schedule U by a series of swaps of nonconflicting operations.

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8. How does the number of page frames affect the number of page faults for a given memory access pattern in FIFO page replacement algorithm?

- (a) Increasing the number of page frames always decreases the number of page faults.
(b) Increasing the number of page frames may increase or decrease the number of page faults depending on the memory access pattern.
(c) Increasing the number of page frames always increases the number of page faults.
(d) Increasing the number of page frames has no effect on the number of page faults.

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9. Match List-I with List-II

List-I		List-II	
A.	Flash memory	I.	Oldest and Slowest
B.	PMOS	II.	Used in large scale integration (LSI)
C.	NMOS	III.	Least power consumption.
D.	CMOS	IV.	Non volatile RAM which is powered continuously.

Choose the correct answer from the options given below

- (a) A-I, B-IV, C-III, D-II (b) A-I, B-IV, C-II, D-III
(c) A-IV, B-I, C-III, D-II (d) A-IV, B-I, C-II, D-III

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ANSWER

1	A	2	B	3	C	4	C	5	D
6	B	7	B	8	B	9	D		

